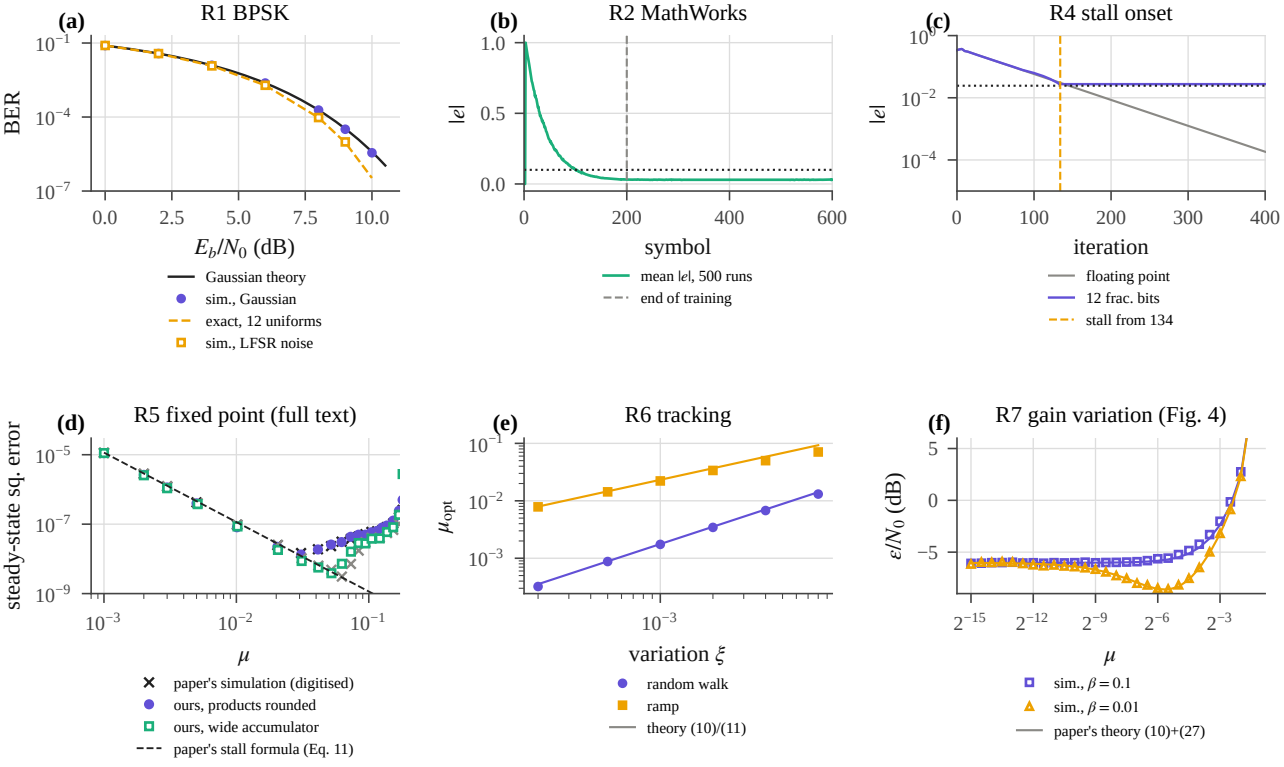
(a)
R1 BPSK
BER
10^{-1}
10^{-4}
10^{-7}
0.0
2.5
5.0
7.5
10.0
E_b/N_0 (dB)
Gaussian theory
sim., Gaussian
exact, 12 uniforms
sim., LFSR noise
(b)
R2 MathWorks
$|e|$
1.0
0.5
0.0
0
200
400
600
symbol
mean |e|, 500 runs
end of training
(c)
R4 stall onset
$|e|$
10^{0}
10^{-2}
10^{-4}
0
100
200
300
400
iteration
floating point
12 frac. bits
stall from 134
(d)
R5 fixed point (full text)
steady-state sq. error
10^{-5}
10^{-7}
10^{-9}
10^{-3}
10^{-2}
10^{-1}
μ
paper's simulation (digitised)
ours, products rounded
ours, wide accumulator
paper's stall formula (Eq. 11)
(e)
R6 tracking
μ_{opt}
10^{-1}
10^{-2}
10^{-3}
10^{-3}
variation ξ
random walk
ramp
theory (10)/(11)
(f)
R7 gain variation (Fig. 4)
ε/N_0 (dB)
5
0
−5
2^{-15}
2^{-12}
2^{-9}
2^{-6}
2^{-3}
μ
sim., $\beta = 0.1$
sim., $\beta = 0.01$
paper's theory (10)+(27)